10

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1. (15 pts) Determine whether the following statements are true or false? No justification is necessary.

(1) If $a_n<0$ for $n>1000$, and $\sum_{n=1}^{\infty} a_n$ converges., then $\sum_{n=1}^{\infty} a_n^2$ may diverge.

False.

$\Rightarrow \sum_{n=1}^{\infty} a_n$ converges, $a_n<0$ for $n>1000$ $\Rightarrow \sum_{n=1}^{\infty}|a_n| = \sum_{n=1}^{1000}|a_n| - \sum_{n=1000}^{1000} a_n$ converges

$\lim_{n\to\infty} a_n$ and $\lim_{n\to\infty} a_n = 0$

$\therefore \lim_{n\to\infty}|a_n| = \lim_{n\to\infty} \frac{a_n^2}{|a_n|} = 0$, $\Sigma|a_n|$ converges. $\Rightarrow \Sigma a_n^2$ converges. False.

(2) Suppose that the power series $\sum_{n=1}^{\infty} a_n(x-1)^n$ converges at $x=0$, and diverges at $x=2$, then the interval of convergence of this series is $[0,2)$.

True

$\Rightarrow$ 1°. $\Sigma a_n(x-1)^n$ converges at $x=0$ $\Rightarrow \Sigma|a_n(x-1)^n$ converges absolutely for $|x-1|<1$

so. $x-1=-1$. i.e $0<x<2$.

$\Rightarrow$

diverges at $x=2$ ie $x-1=1$. $\Rightarrow$ diverges for $|x-1|>1$ $\Rightarrow x>2$ or $x<0$ diverges.

So converges for $0\le x<2$.

Since. $\Sigma a_n x^n$ converges at $x=c\neq 0$ $\Rightarrow$ converges absolutely for all $|x|<|c|$

diverges at $x=d$ $\Rightarrow$ diverges for $|x|>|d|$

or 2°. $\Sigma a_n(x-1)^n$ converges at $x=0$ $\Rightarrow \lim_{n\to\infty} a_n = 0 \Rightarrow \exists \varepsilon=1$ s.t $|a_n|<1$ $\forall n>N$.

$\Rightarrow |a_n(x-1)^n| < |x-1|^n$ for $\forall\ |x-1|<1$

and $\Sigma|x-1|^n$ converges

$\Rightarrow \Sigma|a_n|x-1|^n|$ converges for $|x-1|<1 \Rightarrow 0<x<2$

$\Rightarrow$ converges at $0\le x<2$

3. If $f(x,y)$ has partial derivatives $f_x(x,y)$ and $f_y(x,y)$ at (x_0,y_0), then $\lim_{x\to x_0} f(x,y_0) = \lim_{y\to y_0} f(x_0,y) = f(x_0,y_0)$.

True

$\Rightarrow$ Since $f_x(x_0,y_0) = \lim_{x\to x_0} \frac{f(x,y_0)-f(x_0,y_0)}{x-x_0} = c$, exist, and $x-x_0\to 0 \Rightarrow \lim_{x\to x_0} f(x,y_0) = f(x_0,y_0)$

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$f_y(x_0,y_0) = \lim_{y\to y_0} \frac{f(x_0,y)-f(x_0,y_0)}{y-y_0}$ exist, and $\lim_{y\to y_0}(y-y_0)=0 \Rightarrow \lim_{y\to y_0} f(x_0,y)=f(x_0,y_0)$

$\therefore \lim_{x\to x_0} f(x,y_0) = \lim_{y\to y_0} f(x_0,y) = f(x_0,y_0)$

(4) If a vector function $\vec{r}(t)$ is always perpendicular to its derivative $\frac{d\vec{r}}{dt}$,

then $|\vec{r}(t)|$ must be ~~a function~~ constant.

True

$\Rightarrow$ since $\vec{r}(t)\cdot\frac{d\vec{r}}{dt}=0 \Rightarrow (\vec{r}(t)\cdot\vec{r}(t))' = 2\vec{r}(t)\cdot\frac{d\vec{r}}{dt}=0$

$\Rightarrow |\vec{r}(t)| = c$

(5) The curvature of a unit circle is greater than the curvature of the parabola $y=x^2$ at the origin.

False

$\Rightarrow$ For unit circle: $x^2+y^2=1 \Rightarrow \begin{cases} x=\cos t \\ y=\sin t \end{cases} \Rightarrow \vec{r}(t)=(\cos t)\vec{i}+(\sin t)\vec{j}$

$\frac{d\vec{r}}{dt} = \vec{v}$

$\vec{T} = \vec{v} = (-\sin t)\vec{i}+(\cos t)\vec{j}$

$\left|\frac{d\vec{T}}{dt}\right| = 1$, $|\vec{v}(t)| = 1 \Rightarrow k=1$.

For $y=x^2 \Rightarrow y'=2x$, $y''=2 \Rightarrow y'(0,0)=0$

$$k=\frac{|f''(x)|}{\left(\sqrt{1+(f'(x))^2}\right)^3} = \frac{2}{1} = 2$$

2. Multiple choice Questions: (only one correct answer for each of the following ~~ten~~ questions).

(1) Let $\vec{a}$ and $\vec{b}$ be two nonzero orthogonal vectors, which of the following must be true?

(A) $|\vec{a}+\vec{b}| = |\vec{a}|+|\vec{b}|$ (B) $|\vec{a}-\vec{b}| = |\vec{a}|-|\vec{b}|$

(C) $|\vec{a}+\vec{b}| = |\vec{a}-\vec{b}|$ (D) $\vec{a}+\vec{b} = \vec{a}-\vec{b}$.

$\Rightarrow$ C. $|\vec{a}+\vec{b}| = \sqrt{|\vec{a}|^2+2\vec{a}\cdot\vec{b}+|\vec{b}|^2} = \sqrt{|\vec{a}|^2+|\vec{b}|^2}$, $|\vec{a}-\vec{b}| = \sqrt{|\vec{a}|^2-2\vec{a}\cdot\vec{b}+|\vec{b}|^2} = \sqrt{|\vec{a}|^2+|\vec{b}|^2}$

$\therefore |\vec{a}+\vec{b}| = |\vec{a}-\vec{b}|$.

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(2) The equations of two lines are l_1: $x=t$, $y=2t$, $z=-t$ and l_2: $x=1-2t$, $y=t$, $z=-1+t$. Then l_1 and l_2 are

(A) parallel ; (B) orthogonal ; (C) intersect with each other ; (D) skew

D

$\Rightarrow \vec{v_1}=(1,2,-1)$, $\vec{v_2}=(-2,1,1)$ $\because \vec{v_1}\cdot\vec{v_2}=-1$, $|\vec{v_1}|=|\vec{v_2}|=\sqrt{6}$

$\Rightarrow \cos\theta=-\frac{1}{6}$.

but if it intersect $\begin{cases} x=t=1-2t \\ y=2t=t \end{cases} \Rightarrow \begin{cases} t=\frac{1}{3} \\ t=0 \end{cases}$ ×.

(3) Suppose $0\le a_n<\frac{1}{n}$ $(n=1,2,\cdots)$, then which of the following series must converges?

(A) $\sum_{n=1}^{\infty} a_n$, (B) $\sum_{n=1}^{\infty}(-1)^n a_n$, (C) $\sum_{n=1}^{\infty}\sqrt{a_n}$ (D) $\sum_{n=1}^{\infty}(-1)^n a_n^2$.

D

$\Rightarrow 0\le a_n<\frac{1}{n} \Rightarrow a_n^2<\frac{1}{n^2} \Rightarrow \sum a_n^2$ converges

$\Rightarrow \sum(-1)^n a_n^2$ converges.

3. (9 pts) Does the following series absolutely converge, conditionally converge, or diverge? Give reasons for your answer.

(1) $\sum_{n=1}^{\infty}(-1)^n\left(1-\frac{1}{n}\right)^{n^2}$

$\Rightarrow \lim_{n\to\infty}\sqrt[n]{|a_n|}=\lim_{n\to\infty}\sqrt[n]{(-1)^n\left(1-\frac{1}{n}\right)^{n^2}}=\lim_{n\to\infty}\left(1-\frac{1}{n}\right)^n=e^{-1}<1$

$\Rightarrow$ converges absolutely

(2) $\sum_{n=1}^{\infty}(-1)^n e^{-\frac{1}{n}}$

$\Rightarrow \lim_{n\to\infty} e^{-\frac{1}{n}}=e^0=1\neq 0 \Rightarrow$ diverges.

(3) $\sum_{n=2019}^{\infty}(-1)^n\frac{1}{\sqrt{n^2-2019n+1}}$

$\Rightarrow$ For $\sum_{n=2019}^{\infty}\frac{1}{\sqrt{n^2-2019n+1}}$, $\lim_{n\to\infty}\frac{\frac{1}{\sqrt{n^2-2019n+1}}}{\frac{1}{n}}=1$, $\sum\frac{1}{n}$ diverges $\Rightarrow$ diverges.

$u_n=\frac{1}{\sqrt{n^2-2019n+1}}>0$, $u_n>u_{n+1}$, $u_n\to 0 \Rightarrow$ converges conditionally.

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4. (10 pts)

(1) Find the radius and interval of convergence of the series $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} 2^n x^n}{\sqrt{n^2+n+1}}$

(2) For what values of x does the series converge absolutely, or conditionally?

(1) $\Rightarrow \left|\frac{u_{n+1}}{u_n}\right| = \frac{2^{n+1}\sqrt{n^2+n+1}}{2^n\sqrt{(n+1)^2+(n+1)+1}}|x| \to 2|x|<1 \Rightarrow |x|<\frac{1}{2} \Rightarrow \boxed{R=\frac{1}{2}}$

(2) when $x=\frac{1}{2}$ $\sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n^2+n+1}}$ converges conditionally since $\frac{1}{\sqrt{n^2+n+1}} \neq 0$, $\lim_{n\to\infty} \frac{1}{\sqrt{n^2+n+1}} = 0$,

$\lim_{n\to\infty} \frac{\frac{1}{\sqrt{n^2+n+1}}}{\frac{1}{n}} = 1$, $\sum \frac{1}{n}$ diverges.

$\Rightarrow \sum \frac{1}{\sqrt{n^2+n+1}}$ diverges.

when $x=-\frac{1}{2}$, $\sum_{n=0}^{\infty} \frac{(-1)^{n+1}\cdot(-1)^n}{\sqrt{n^2+n+1}} = \sum_{n=0}^{\infty} \frac{-1}{\sqrt{n^2+n+1}}$ diverges since $\lim_{n\to\infty} \frac{\frac{1}{\sqrt{n^2+n+1}}}{\frac{1}{n}} = 1$, $\sum \frac{1}{n}$ diverges.

$\Rightarrow$ converges absolutely: $(-\frac{1}{2}, \frac{1}{2})$; converges: $(-\frac{1}{2}, \frac{1}{2}]$

converges conditionally: $x=\frac{1}{2}$.

5. (10 pts)

Let $x=\cos^3 t$, $y=\sin^3 t$, where $0\le t\le \frac{\pi}{2}$, be a parametrization of a curve.

(1) Find the length of the curve.

(2) Find the area of the surface generated by revolving the curve about the x-axis.

$\Rightarrow$ (1) $\frac{dx}{dt} = -3\cos^2 t \sin t$, $\frac{dy}{dt} = 3\sin^2 t \cos t$ $\Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9\sin^2 t\cos^2 t$

$$L = \int_0^{\frac{\pi}{2}} \sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt = \int_0^{\frac{\pi}{2}} \sqrt{9\sin^2 t\cos^2 t}\,dt = \int_0^{\frac{\pi}{2}} 3\sin t\cos t\,dt = \int_0^{\frac{\pi}{2}} 3\sin t\,d\sin t$$

$$= \frac{3}{2}\sin^2 t\Big]_0^{\frac{\pi}{2}} = \boxed{\frac{3}{2}}$$

(2) $$L = \int_0^{\frac{\pi}{2}} 2\pi y\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt = \int_0^{\frac{\pi}{2}} 2\pi\sin^3 t\cdot 3\sin t\cos t\,dt = \int_0^{\frac{\pi}{2}} 6\pi\sin^4 t\,d\sin t.$$

$$= \frac{6}{5}\pi\sin^5 t\Big]_0^{\frac{\pi}{2}} = \boxed{\frac{6}{5}\pi}$$

6. (10pts) Find the equation of the plane through the points $(2,1,-1)$ and $(1,0,$ perpendicular to the plane $2x+3y-5z+6=0$.

$\Rightarrow$ Let $P(2,1,-1)$, $Q(1,0,-1) \Rightarrow \vec{PQ} = (-1,1,0)$, $\vec{n_1} = (2,3,-5)$

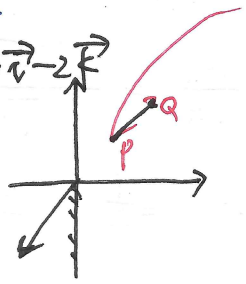
$$\Rightarrow \vec{R} = \vec{PQ}\times\vec{n_1} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 1 & 0 \\ 2 & 3 & -5 \end{vmatrix} = -5\vec{i} - 5\vec{j} - 5\vec{k}$$

$\Rightarrow$ the plane: $-5(x-2)-5(x+1)-5(z+$

$\Rightarrow \boxed{x+y+z=0.}$

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7. (10 pts) A particle is located at the point $(1, 0, -2)$. Its initial speed is $|\vec{v}(0)| = 3$ at time $t=0$, and the direction of its initial velocity is toward the point $(2, -1, 3)$. The particle moves with constant acceleration $\vec{i}+2\vec{j}+\vec{k}$. Find its position vector $\vec{r}(t)$ at time t.



$\Rightarrow$ let $P(1,0,-2)$, $Q(2,-1,3)$ $\Rightarrow \overrightarrow{PQ} = (1,-1,5)$, $\vec{r}(0) = \vec{i} - 2\vec{k}$

the direction of $\vec{v}(0)$ is $\frac{\overrightarrow{PQ}}{|\overrightarrow{PQ}|} = \frac{1}{\sqrt{27}}(1,-1,5)$

so $\vec{v}(0) = |\vec{v}(0)| \cdot \frac{\overrightarrow{PQ}}{|\overrightarrow{PQ}|} = 3 \cdot \frac{1}{3\sqrt{3}}(1,-1,5) = \frac{1}{\sqrt{3}}(1,-1,5)$

$\vec{a}(t) = \vec{i} + 2\vec{j} + \vec{k}$

$\Rightarrow \vec{v}(t) = \int (\vec{i}+2\vec{j}+\vec{k})\,dt = t\vec{i} + 2t\vec{j} + t\vec{k} + \vec{C}$

$\vec{v}(0) = \frac{1}{\sqrt{3}}\vec{i} - \frac{1}{\sqrt{3}}\vec{j} + \frac{5}{\sqrt{3}}\vec{k}$

$\rightarrow \vec{C} = \frac{1}{\sqrt{3}}\vec{i} - \frac{1}{\sqrt{3}}\vec{j} + \frac{5}{\sqrt{3}}\vec{k}$

$\rightarrow \vec{v}(t) = (t+\frac{1}{\sqrt{3}})\vec{i} + (2t - \frac{1}{\sqrt{3}})\vec{j} + (t + \frac{5}{\sqrt{3}})\vec{k}$

$\vec{r}(t) = \int [(t+\frac{1}{\sqrt{3}})\vec{i} + (2t - \frac{1}{\sqrt{3}})\vec{j} + (t+\frac{5}{\sqrt{3}})\vec{k}]\,dt$

$= (\frac{1}{2}t^2 + \frac{1}{\sqrt{3}}t)\vec{i} + (t^2 - \frac{1}{\sqrt{3}}t)\vec{j} + (\frac{1}{2}t^2 + \frac{5}{\sqrt{3}}t)\vec{k} + \vec{C}$

$\vec{r}(0) = \vec{i} - 2\vec{k} \Rightarrow \vec{C} = \vec{r}(0) = \vec{i} - 2\vec{k}$

$\therefore \vec{r}(t) = (\frac{1}{2}t^2 + \frac{1}{\sqrt{3}}t + 1)\vec{i} + (t^2 - \frac{1}{\sqrt{3}}t)\vec{j} + (\frac{1}{2}t^2 + \frac{5}{\sqrt{3}}t - 2)\vec{k}$.

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8. (10 pts)

Is the following function, $f(x,y)$ continuous at $(0,0)$? Give reasons for your answer.

$$f(x,y)=\begin{cases}\dfrac{\sin(x^3+y^3)}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$$

$$\Rightarrow \lim_{(x,y)\to(0,0)} f(x,y)=\lim_{(x,y)\to(0,0)}\frac{\sin(x^3+y^3)}{x^2+y^2}=\lim_{(x,y)\to(0,0)}\frac{\sin(x^3+y^3)}{x^3+y^3}\cdot\frac{x^3+y^3}{x^2+y^2}=\lim_{(x,y)\to(0,0)}\frac{x^3+y^3}{x^2+y^2}$$

$$=\lim_{r\to 0}\frac{r^3\cos^3\theta+r^2\sin^3\theta}{r^2}=\lim_{r\to 0} r(\cos^3\theta+\sin^3\theta)=0.$$

or $$0\le\left|\frac{x^3+y^3}{x^2+y^2}\right|=\left|\frac{x\cdot x^2}{x^2+y^2}+\frac{y\cdot y^2}{x^2+y^2}\right|=\left|x\cdot\frac{x^2}{x^2+y^2}+y\cdot\frac{y^2}{x^2+y^2}\right|\le|x|+|y|$$

$$\lim_{(x,y)\to(0,0)}(|x|+|y|)=0$$

$$\Rightarrow \lim_{(x,y)\to(0,0)}\left|\frac{x^3+y^3}{x^2+y^2}\right|=0 \Rightarrow \lim_{(x,y)\to(0,0)}\frac{x^3+y^3}{x^2+y^2}=0$$

$$\Rightarrow \lim_{(x,y)\to(0,0)} f(x,y)=0=f(0,0)$$

$f(x,y)$ is continuous at $(0,0)$.

9. (9 pts) Let $f(u)$ be differentiable, $z=f(e^x y)$ $(y\neq 0)$, and $\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=1$.

If $f(1)=0$, find $f(u)$.

$$\Rightarrow \frac{\partial z}{\partial x}=f'(e^xy)\cdot e^x y,\quad \frac{\partial z}{\partial y}=f'(e^xy)e^x$$

$$\Rightarrow \frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=2f'(e^xy)e^xy=1,\ \text{let } u=e^xy$$

$$\Rightarrow 2f'(u)u=1 \Rightarrow \frac{df}{du}=\frac{1}{2u} \Rightarrow f(u)=\frac{1}{2}\ln|u|+c$$

$$f(1)=c=0 \Rightarrow \boxed{f(u)=\frac{1}{2}\ln|u|}$$

or $=\frac{1}{2}\ln|2u|+c$

$f(1)=\frac{1}{2}\ln 2+c_1=0$

$c_1=-\frac{1}{2}\ln 2$

$\Rightarrow f(u)=\frac{1}{2}\ln|u|-\frac{1}{2}\ln 2$

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10. (8 pts)

Find the Taylor series for $f(x)=\ln(x+\sqrt{x^2+1})$ at $x=0$.

$$\Rightarrow f'(x)=\frac{1+\frac{x}{\sqrt{x^2+1}}}{x+\sqrt{x^2+1}}=\frac{1}{\sqrt{x^2+1}}=(1+x^2)^{-\frac{1}{2}}=1+\sum_{k=1}^{\infty}\binom{-\frac{1}{2}}{k}x^{2k},\ -1<x<1$$

$$\Rightarrow f(x)=x+\sum_{k=1}^{\infty}\frac{1}{2k+1}\binom{-\frac{1}{2}}{k}x^{2k+1},\ -1<x<1.$$

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